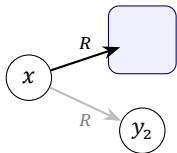


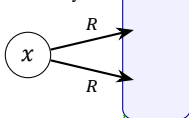
$\exists R.C$
 “at least one R -link to C ”



$x \in (\exists R.C)^I \quad \square$

There exists $y_1 \in C^I$ with
 $(x, y_1) \in R^I$

$\forall R.C_{C^I}$
 “every R -link leads to C ”



$\square \text{ all } \in C^I$



no R -link

$z \in (\forall R.C)^I$ (vacuous)

Distinction: Absence in RDF data $\neq$ absence in an interpretation.
 Under OWA, a missing R triple for z does NOT entail $z : \forall R.C$.